llmXive Automated Review of A Stylometric Application of Large Language Models

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper generally makes accurate claims supported by the provided data and statistical tests, with one notable exception regarding the interpretation of a p-value in the ablation studies.

In the "Ablation studies" section (Results), the authors claim that models trained on part-of-speech (POS) corpora were "significantly less effective" than those trained on function-word-only corpora, citing a t-statistic of 2.11 and a p-value of 6.04×10^{-2} (0.0604). By standard scientific convention ($\alpha < 0.05$), a p-value of 0.0604 does not support a claim of statistical significance. The text should be corrected to reflect that the difference was not statistically significant at the 0.05 level, or described as "marginally significant" to avoid misleading the reader about the strength of the evidence.

Additionally, the Abstract and Introduction state that the study confirms R. P. Thompson's authorship of the 15th Oz book, which is "now the accepted attribution." While this is largely true following the work of Binongo (2003), the phrasing implies a pre-existing consensus that the current study merely confirms. It would be more accurate to frame this as confirming the findings of Binongo (2003) and resolving a historical attribution debate, rather than confirming a settled fact. The citation to Binongo (2003) is present, but the narrative framing slightly overstates the prior state of consensus.

Finally, in the ablation results, the claim that content-word models were "significantly less effective" than intact texts is supported by a t-test with non-integer degrees of freedom ($t(11.77)$). While this likely indicates a Welch's t-test was performed, the manuscript does not explicitly state this. For full transparency and accuracy in reporting statistical methods, the specific test used (e.g., Welch's t-test) should be named to justify the degrees of freedom and the resulting p-value.

Action items.

- [**writing**] The paper generally makes accurate claims supported by the provided data and statistical tests, with one notable exception regarding the interpretation of a p-value in the ablation studies. In the "Ablation studies" section (Results), the authors claim that models trained on part-of-speech (POS) corpora were "significantly less effective" than those trained on function-word-only corpora, citing a t-statistic of 2.11 and a p-value of 6.04×10^{-2} (0.0604). By standard scientific conventi

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1A contains a discrepancy between the legend and the plotted lines in the 'Train' subplot, and the caption's description of color mapping is inconsistent with the visual data where the training author's curve is not the lowest.

Figure 2

The figure is visually clear with appropriate axes and legends, but the caption contains a repeated word typo, is truncated at the end, and the author count in the legend (9) conflicts with the caption's mention of 'eight authors'.

Figure 3

The figure is visually clear but contains a scientific contradiction where the diagonal values are non-zero despite the caption claiming baseline subtraction. Additionally, the caption has a broken reference to a supplementary figure.

Figure 4

The figure is a 3D scatter plot of author names but lacks axis labels and scales, making the data uninterpretable. Additionally, the caption contains broken cross-references with missing figure numbers.

Figure 5

The figure effectively visualizes the loss curves for Baum and Thompson models across different datasets, but the caption is severely truncated and contains a broken figure reference, hindering full comprehension.

Figure 6

The figure follows the format of the main text but suffers from a truncated caption that cuts off the description of the scatter plot points. Additionally, the x-axis in Panel B lacks a label identifying the 'Training author'.

Figure 7

The figure follows the standard format with clear axes and legends, but the caption is truncated mid-sentence, and the x-axis label is missing from the upper rows of Panel A.

Figure 8

The figure panels are clear and follow the established format, but the caption is truncated at the end and contains a broken cross-reference to the main text figure.

Figure 9

The figure is visually clear with a complete legend and axes, but the caption contains typos, is truncated, and incorrectly describes a black curve in Panel B that is missing from the plot.

Figure 10

The figure is visually clear and follows the format of the main text, but the caption contains a truncation error, a typo, and a minor mismatch regarding the plurality of the black threshold line.

Figure 11

The figure caption contains a filename mismatch and is truncated, while the plot in Panel A appears to show the average t-statistic (similar to Panel B) rather than the individual author curves described in the text.

Figure 12

The figure presents clear heatmaps, but the caption's claim that baseline loss was subtracted is contradicted by the non-zero diagonal values shown in the matrices. Additionally, the filename in the caption contains a spelling error.

Action items.

- [**science**] Figure 1A: The caption states 'Each color denotes a model trained on a single author's work,' but the legend lists 8 authors while the 'Train' subplot contains 9 distinct colored lines (including a cyan line not in the legend). Additionally, the 'Train' subplot shows a line for the training author (e.g., green for Austen) that is not the lowest loss curve, contradicting the expectation that a model should fit its own training data best.
- [**writing**] Figure 1A: The 'Train' subplot lacks a corresponding legend entry for the cyan line visible in the plot, creating ambiguity about which author/model it represents.
- [**writing**] Figure 2 caption contains a typo: 'the the *t*-statistic' (repeated word).
- [**writing**] Figure 2 caption is truncated mid-sentence at the end ('...only content wor').
- [**science**] Figure 2A: The legend lists nine authors (Baum, Thompson, Austen, Dickens, Fitzgerald, Melville, Twain, Wells) but the caption states comparisons are across 'eight authors'; verify count consistency.
- [**science**] Figure 3: The caption states the matrix displays loss 'after subtracting the native author's baseline loss,' which implies the diagonal values (native author vs. native author) should be zero. However, the diagonal contains non-zero values (e.g., 3.52, 3.43, 3.64), contradicting the description.
- [**writing**] Figure 3: The caption contains a broken cross-reference ('Supp. Fig. ') with no figure number provided.

- **[fatal]** Figure 4: The caption contains a broken cross-reference (‘shown in Figure .’) with a missing figure number, making it impossible to verify the data source.
- **[science]** Figure 4: The 3D plot lacks labeled axes with units or scales, rendering the spatial coordinates and distances between authors uninterpretable.
- **[writing]** Figure 4: The caption contains a broken reference to supplementary materials (‘Supp. Fig. .’) with a missing figure number.
- **[writing]** Figure 5: The caption contains a broken cross-reference (‘from Figure —‘) and ends abruptly mid-sentence (‘...Baum-trained models;’), indicating incomplete text.
- **[writing]** Figure 5: The top-left subplot is titled ‘Training’ but the caption does not explicitly describe this panel, creating a disconnect between the visual label and the textual description.
- **[writing]** Figure 6: The caption text is truncated at the end (‘Each dot [loss_all_authors_content_only.pdf]’), cutting off the description of the data points.
- **[writing]** Figure 6: The x-axis label ‘Training author’ in Panel B is missing, whereas the caption explicitly states the x-axis represents the model’s training author.
- **[writing]** Figure 7 caption is truncated mid-sentence at the end (‘Each [loss_all_authors_function_only.pdf]’), cutting off the description of the data points in Panel B.
- **[writing]** Figure 7 Panel A: The x-axis label ‘Epochs completed’ is present only on the bottom row of plots; it is missing from the top two rows, reducing clarity for those subplots.
- **[writing]** Figure 8 caption is truncated mid-sentence at the end (‘...or from other [loss_all_authors_pos.pdf]’), cutting off the description of the data points in Panel B.
- **[writing]** Figure 8 caption contains a broken cross-reference (‘Figure in the main text’) instead of specifying the figure number (e.g., Figure 1).
- **[writing]** Figure 9 caption contains a typo: ‘the the t-statistic’ (repeated word).
- **[writing]** Figure 9 caption is truncated mid-sentence at the end (‘...for each ep’).
- **[science]** Figure 9 caption states ‘The black curves in both panels’ (plural), but only Panel A contains a black curve; Panel B lacks the described significance threshold line.
- **[science]** Figure 9 caption claims ‘All function words are masked out using <FUNC>’, but the rendered plot contains no visual indication (e.g., label, note) that this specific preprocessing step was applied, making the figure indistinguishable from the unmasked version without external context.
- **[writing]** Figure 10: The caption text is truncated at the end (‘for eac’), cutting off the description of the black curves and the file reference.
- **[writing]** Figure 10: The caption contains a typo (‘the the’) in the description of panel A.
- **[science]** Figure 10: Panel A displays a single black line representing the p=0.001 threshold, but the caption states ‘The black curves’ (plural), creating a mismatch between the visual and the text.

- **[fatal]** Figure 11: The caption filename '[t_stats_content_only.pdf]' contradicts the title 'using only parts of speech' and the format of Figure 10 (function words); likely a copy-paste error.
- **[fatal]** Figure 11: The caption text is truncated at the end ('...corresponding to'), missing the significance threshold value (e.g., p=0.001) referenced by the black curves.
- **[science]** Figure 11: Panel A displays a single black curve and a gray ribbon but lacks the multi-colored author curves described in the caption ('Each curve denotes... the given author (color)'); the plot appears to show the average (Panel B format) instead of individual authors.
- **[science]** Figure 12: The caption states the matrices display loss 'after subtracting the native author's baseline loss,' yet the diagonal values (e.g., Baum-Baum in Panel A is 2.76) are non-zero positive numbers. If baseline subtraction were applied, the diagonal should be 0.00. This contradicts the caption's description of the data processing.
- **[writing]** Figure 12: The caption contains a typo in the filename reference ('confustion_matrices_variants.pdf' instead of 'confusion...').

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript demonstrates a strong command of computational linguistics terminology but occasionally relies on jargon that may alienate readers from the digital humanities or general literary studies, who are a key audience for this work.

First, the term **"predictive comparison"** is introduced in the Introduction (Section 1) as a "new LLM-based relative stylometric measure." While the authors explain the *idea* (training a model on one author and testing on another), they do not explicitly define the *term* itself in plain language. A non-specialist might struggle to distinguish this from standard "perplexity" or "cross-entropy" metrics without a clearer, jargon-free definition of the specific comparative mechanism.

Second, the phrase **"stylometric signatures"** appears frequently (e.g., Section 2.1, 3.1, 3.4). In a biological or forensic context, a "signature" implies a unique, immutable identifier. In this statistical context, the authors are measuring probabilistic patterns. Using "stylistic patterns," "writing fingerprints," or "statistical profiles" would be more accurate and less likely to mislead readers about the nature of the evidence.

Third, the term **"ablation studies"** is used in Section 3.4 to describe the experiments where content words, function words, or parts of speech are removed. While standard in machine learning, this term is not widely known outside the field. A brief parenthetical explanation (e.g., "experiments where specific components are removed to test their contribution") would make the methodology accessible to literary scholars.

Fourth, in the Discussion (Section 4.3), the authors suggest using **"parameter-efficient fine-tuning"** methods. This is a specific technical concept (often referring to techniques like LoRA or adapters) that is not defined. For a general audience, this should be rephrased as "methods that update only a small part of the model" or "lightweight adaptation techniques."

Finally, the word **"confounds"** in Section 2.1 ("eliminating any potential confounds due to translation") is a statistical shorthand. Replacing it with "confounding factors" or "interfering variables" would improve clarity for non-statisticians.

Addressing these terms will ensure the paper's insights are accessible to the interdisciplinary audience it aims to reach, without sacrificing technical precision.

Action items.

- [**writing**] The manuscript demonstrates a strong command of computational linguistics terminology but occasionally relies on jargon that may alienate readers from the digital humanities or general literary studies, who are a key audience for this work. First, the term "predictive comparison" is introduced in the Introduction (Section 1) as a "new LLM-based relative stylometric measure." While the authors explain the *idea* (training a model on one author and testing on another), they do not explicitly defin

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a logically coherent argument that LLM cross-entropy loss can serve as a stylometric measure. The core premise—that a model trained on Author A will predict Author A's text better than Author B's—is consistently applied. However, three specific logical gaps require tightening to ensure conclusions strictly follow from premises.

First, the claim of "perfect (100%) classification accuracy" in Section 3.1 rests on the assertion that the same-author model *always* yields the lowest loss. While Figure 1B supports this for the full-text models, the ablation studies in the Supplementary Materials reveal that for specific linguistic subsets (e.g., Melville with content words, Austen with function words), the t-statistics are not significant ($p > 0.05$), indicating overlapping loss distributions. If distributions overlap, the "always" condition is technically violated for those subsets. While the main claim refers to full-text models, the logical rigor would improve by explicitly stating that the "perfect" separation is a property of the full-text models specifically, distinguishing it from the ablated conditions where separation is probabilistic rather than absolute.

Second, the definition of stylometric distance $d(i, j)$ in Section 3.2 involves symmetrizing the asymmetric cross-entropy loss matrix ($L_j(i) \neq L_i(j)$). The authors define $d(i, j)$ as the average of the normalized losses in both directions and use this to generate an MDS plot. MDS typically assumes a symmetric distance metric. The paper does not explicitly justify why the asymmetry of the underlying cross-entropy loss does not invalidate the geometric interpretation of the resulting "distance" or the MDS projection. A brief logical bridge explaining why the average is a sufficient proxy for a symmetric distance in this context would strengthen the argument.

Finally, in Section 3.4, the authors conclude that grammatical structure (POS) is "less distinctive" because POS-only models failed to distinguish authors. However, the text simultaneously states these models "rapidly converged to low training and evaluation losses." Logically, convergence implies the model successfully learned the statistical regularities of the POS sequences. The failure to distinguish authors implies those regularities are similar across authors, not that the model failed to learn them. The current phrasing risks conflating "the model failed to learn" with "the feature is not distinctive." Clarifying that the models *did* learn the POS patterns, but that these patterns were insufficient for discrimination, would make the causal chain from convergence to lack of distinctiveness more rigorous.

Action items.

- [**science**] The claim of 'perfect (100%) classification accuracy' (Sec 3.1) relies on the premise that the same-author model always yields the lowest loss. However, ablation results (Supp. Tab 1) show non-significant t-stats for some subsets (e.g., Melville content-only), implying loss distribution overlap. Explicitly confirm the full model's separation is absolute across all seeds to support the 'always' claim.
- [**writing**] The stylometric distance d(i,j) in Sec 3.2 symmetrizes asymmetric cross-entropy losses. The text assumes this average accurately reflects 'distance' but lacks justification for why the inherent asymmetry of cross-entropy does not distort the MDS projection, which assumes symmetric distances. Clarify this logical step.
- [**writing**] In Sec 3.4, the conclusion that POS structure is 'less distinctive' follows from POS-only models failing to distinguish authors. However, the text notes these models 'rapidly converged,' implying they learned the patterns. The logic conflates 'learning capability' with 'distinctiveness.' Clarify that the models learned the patterns, but those patterns were insufficient for discrimination.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several claims that extend beyond the immediate scope of the experimental data presented. While the core finding—that GPT-2 models trained on single authors distinguish their own held-out text from others with high accuracy—is well-supported by the provided t-tests and confusion matrices, the interpretation of these results occasionally overstates the implications.

First, the Abstract and Introduction repeatedly assert that the trained models "embody the unique writing style" of the author (Abstract, p.1; Intro, p.2). The data demonstrates that the models capture *discriminative* statistical patterns, but “uniqueness” is a strong claim not fully justified by a dataset of only eight authors from a specific historical period and genre (classic English literature). The models may be capturing genre-specific or era-specific markers rather than truly unique authorial signatures. The language should be tempered to reflect that the models capture “author-specific statistical regularities” rather than “unique style.”

Second, the paper claims to “confirm R. P. Thompson’s authorship” of the 15th Oz book (Abstract, p.1; Intro, p.2). This is a logical overreach. The study validates the *method* by showing it correctly classifies a text with *known* ground truth (Thompson). It does not provide new evidence to “confirm” the attribution in a historical or literary sense; it merely replicates a known result. The phrasing should be adjusted to state that the method “successfully recovers the accepted attribution” or “is consistent with the established attribution,” avoiding the implication that the paper itself is the source of confirmation.

Finally, the Discussion section suggests the approach “naturally extends to open-set attribution problems” where new authors can be introduced without retraining existing models (Discussion, p.10). This is an overstatement of the current capabilities. The proposed method requires training a separate model from scratch for *each* author. In a true open-set scenario with thousands of potential authors, the computational cost of training a new model for every candidate would be prohibitive. The paper does not demonstrate this scalability or offer a mechanism for efficient open-set inference. This claim should be qualified as a “theoretical direction” or “future work” rather than a current feature of the method.

Action items.

- [**writing**] The claim that the method ‘embodies the unique writing style’ (Abstract, p.1) over-interprets the results. The data shows the model captures statistical regularities sufficient for discrimination, but ‘uniqueness’ implies a level of distinctiveness not proven against a larger, more diverse author pool. Temper this to ‘author-specific statistical patterns’.
- [**writing**] The statement that the approach ‘confirms R. P. Thompson’s authorship’ (Abstract, p.1) and ‘confirming what is now the accepted attribution’ (Intro, p.2) is circular. The paper validates the method against a known ground truth but does not provide new evidence to ‘confirm’ the attribution itself. Rephrase to state the method ‘successfully recovers the accepted attribution’.
- [**science**] The claim that the method ‘naturally extends to open-set attribution problems’ (Discussion, p.10) is an overreach. The current experiment is a closed-set classification task (8 authors). The computational cost of training a new model for every potential new author in an open set is prohibitive, and the paper does not demonstrate this scalability. Qualify this as a ‘theoretical possibility’ rather than a demonstrated extension.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper addresses safety and ethics primarily through the lens of authorship attribution on public domain texts, which minimizes immediate privacy and consent concerns. The dataset consists entirely of works from Project Gutenberg (Section 2.1, lines 65-70), ensuring that no personal data or copyrighted material from living authors is used without permission. This is a strong point for the current scope.

However, there are two specific areas requiring attention regarding ethical standards and potential dual-use risks:

1. **Authorship Attribution of AI:** The title block (line 24) lists `qwen.qwen3.5-122b` as a co-author alongside human researchers. This violates standard academic ethical guidelines (such as those from the Committee on Publication Ethics or the ACL Code of Ethics), which state that AI systems cannot be authors as they cannot take responsibility for the work. The LLM's contribution should be moved to the Acknowledgments section (Section 6.1), and the author list should be corrected to reflect only human contributors.
2. **Dual-Use and Future Misuse:** While the current experiments are benign (analyzing 19th/early 20th-century literature), the Discussion section (lines 430-440) explicitly proposes future applications for generating "counterfactual texts" in the style of authors, potentially including modern ones. The paper currently lacks a discussion on the risks of this technology being used for impersonation, creating non-consensual deepfake text, or bypassing copyright protections for living authors. Given the paper's focus on "predictive comparison" as a tool for attribution, it is crucial to acknowledge that the same mechanism could be used to forge authorship or deceive readers. A brief paragraph in the Discussion or Limitations section addressing these potential misuse cases and the ethical boundaries of applying this method to contemporary works is recommended.

No IRB or IACUC concerns are identified as the data is public and non-human. The primary ethical gaps are the improper listing of an AI as an author and the lack of a risk assessment for future applications involving living authors.

Action items.

- **[writing]** The manuscript lists an LLM (qwen.qwen3.5-122b) as a co-author in the title block (line 24). Current ethical guidelines (e.g., COPE, ACL) generally prohibit listing AI systems as authors. This should be corrected to acknowledge the LLM's contribution in the Acknowledgments section instead.
- **[writing]** The paper discusses authorship attribution for historical texts (public domain). However, the 'Discussion' section (lines 430-440) explicitly suggests future applications for generating 'counterfactual texts' in the style of living or modern authors (e.g., Austen on social media). The authors should add a brief ethical caveat regarding the potential misuse of such tools for impersonation, deepfakes, or copyright infringement if applied to contemporary works.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling application of LLM cross-entropy loss for stylometry, demonstrating perfect classification accuracy on a small set of eight authors. The experimental design, utilizing 10 random seeds for data sampling and reporting bootstrap confidence intervals, generally supports the robustness of the central claim that LLMs capture author-specific styles. However, several aspects of the evidence require clarification to rule out alternative explanations and ensure statistical rigor.

First, the ablation studies in the Supplementary Materials reveal specific failures that are glossed over in the main text. Table 2 (content words) shows Melville's model fails to distinguish his own text from others ($p = 0.2274$), and Table 3 (function words) shows Austen's model fails ($p = 0.6581$). The text states models "reliably learned" patterns for 6/8 and 5/8 authors, respectively, but does not analyze *why* these specific authors failed. Given the small sample size (8 authors), these failures could indicate that the stylometric signal for these authors is not robust to the specific ablation method, or that their corpora (e.g., Melville's varied genres) introduce noise that the ablation exacerbates. A discussion of these specific outliers is necessary to validate the generalizability of the ablation conclusions.

Second, the experimental design regarding the held-out book introduces a potential confound. The methods state: "we randomly select one book to hold out for evaluation." It is unclear if this selection was randomized across the 10 seeds or if a single book was held out for all seeds. If a single book was held out, the results may be driven by the specific stylistic or thematic properties of that single book rather than the author's general style. For instance, if the held-out book is a short story while the training set is novels, the model might be distinguishing genre rather than author. Randomizing the held-out book across seeds would strengthen the evidence that the models capture general authorial signatures.

Finally, the training protocol uses a fixed loss threshold (3.0) as the stopping criterion. While the authors justify this by stating it allows for "fair comparison," it introduces a variable training duration. Models that reach 3.0 quickly may be under-trained compared to those that require more epochs, or conversely, models that struggle to reach 3.0 might be overfitting to the training distribution. The reliance on "manual inspection" of generated samples to set this threshold is subjective. A more rigorous approach would involve monitoring validation loss on a held-out set of the *same* author to prevent overfitting, or training all models for a fixed number of epochs to ensure equal capacity, then comparing the resulting losses. The current method risks conflating training dynamics with stylistic distinctiveness.

Action items.

- **[science]** The ablation study results for Melville (content words) and Austen (function words) show non-significant p-values (0.2274 and 0.6581 respectively, Supp. Tab. 2 & 3). The text claims 'reliable' learning for 6/8 and 5/8 authors but does not explicitly discuss these specific failures or potential confounds (e.g., corpus size, genre homogeneity) that might explain the lack of signal for these specific authors.

- [**science**] The study relies on a single random seed for the held-out book selection per author (one book held out, rest used for training). While 10 seeds are used for sampling sub-sequences, the specific choice of the held-out book is not randomized across the full corpus. This introduces a potential confound where the held-out book's specific topic or era might drive the results rather than general authorial style.
- [**science**] The training stopping criterion is a fixed loss threshold (3.0) rather than a fixed number of epochs or early stopping based on validation loss. This creates a risk that models for different authors converge to different effective capacities or overfit to different degrees, potentially biasing the cross-entropy comparisons. The justification ('manual inspection') is anecdotal and lacks statistical rigor.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis presented in the manuscript is generally robust in its experimental design, particularly the use of 10 random seeds to generate distributions for hypothesis testing. The core finding—that models trained on a specific author yield significantly lower cross-entropy loss on that author's held-out text compared to others—is supported by strong effect sizes and highly significant p-values in the primary analysis (Table 1).

However, several statistical reporting and inference details require clarification and correction before the results can be fully accepted.

First, the degrees of freedom (df) reported in Table 1 and the Supplementary Tables are non-integers (e.g., df = 31.53 for Baum, df = 16.39 for Thompson). This indicates the authors utilized Welch's t-test (which does not assume equal variances) rather than the standard Student's t-test. While this is often the more appropriate choice, the manuscript text (Section "Results") does not explicitly state that Welch's t-test was used. The methods section should be updated to specify the exact statistical test variant employed to ensure reproducibility and clarity regarding the assumption of variance homogeneity.

Second, the manuscript does not address the issue of multiple comparisons. The authors conduct a series of hypothesis tests: 8 primary tests for the main authors, and then 3 additional sets of 8 tests for the ablation studies (content-only, function-only, POS-only), totaling 32 tests. The reported p-values are uncorrected. For the ablation studies, where some results are marginal (e.g., Melville in the function-word model, p = 0.0529 in Supp. Tab. 2), the lack of correction (such as Bonferroni or Benjamini-Hochberg) is a significant concern. The authors should apply a correction method to the ablation comparisons and report the adjusted p-values to confirm that the conclusions regarding the relative effectiveness of content vs. function words remain valid.

Finally, the statistical power of the analysis relies on n=10 (the number of random seeds). While sufficient for the main effect (which shows massive separation), the power is limited for the ablation studies where effect sizes are smaller. The discussion of the POS-only results (where the average t-value was not reliably greater than zero, p=0.141) would benefit from a brief comment on the stability of these null results given the small sample size. The authors should ensure that the claim that "POS sequences... appear to be more similar across authors" is not an artifact of low power, although the trend is consistent across the data.

Overall, the statistical approach is sound, but the reporting of test assumptions and the handling of multiple comparisons need to be tightened to meet rigorous standards.

Action items.

- [**writing**] Clarify the degrees of freedom (df) reported in Table 1 and Supplementary Tables. The df values are non-integers (e.g., 31.53, 16.39), implying Welch's t-test was used, but the text does not explicitly state this assumption or justify the variance inequality. Explicitly name the test variant used.
- [**science**] Address the multiple comparisons problem. The study performs 8 primary t-tests (one per author) and 3 additional sets of 8 tests for ablation studies (24 total). The text reports uncorrected p-values (e.g., $p < 0.001$). Apply a correction method (e.g., Bonferroni or Benjamini-Hochberg) to the ablation comparisons to ensure the reported significance holds.
- [**science**] Justify the use of 10 random seeds as the basis for statistical inference. With n=10, the power to detect effect sizes in the ablation studies (where some p-values are marginal, e.g., p=0.0529 for Melville in function-word models) is low. Discuss the stability of these results or provide power analysis estimates.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript is generally well-written, with clear and concise prose that effectively communicates the research findings. The flow of ideas is logical, and the structure of the paper is well-organized. However, there are several instances of a recurring typo in the supplementary materials that need to be addressed.

Specifically, in the captions of Figures 2 through 100 in `supplement.tex`, the phrase "the the t-statistic" appears repeatedly. This is a clear grammatical error that should be corrected to "the t-statistic" for consistency and clarity. While this is a minor issue, it affects the overall professionalism of the document and should be fixed before publication.

Additionally, the main text is free of significant grammatical errors, and the sentences are well-constructed. The use of technical terms is appropriate, and the explanations are clear. The abstract and introduction provide a good overview of the research, and the methods and results sections are detailed and easy to follow.

In summary, the writing quality is high, but the recurring typo in the supplementary materials needs to be corrected. Once this issue is addressed, the manuscript will be ready for publication.

Action items.

- [**writing**] In the caption of Figure 2 (supplement.tex), the phrase 'the the t-statistic' contains a repeated article. Please remove the duplicate 'the' to correct the grammatical error.
- [**writing**] In the caption of Figure 6 (supplement.tex), the phrase 'the the t-statistic' appears again. This is a recurring typo that should be fixed for consistency and clarity.
- [**writing**] In the caption of Figure 7 (supplement.tex), the phrase 'the the t-statistic' is repeated. Please correct this typo to ensure professional quality in the supplementary materials.
- [**writing**] In the caption of Figure 8 (supplement.tex), the phrase 'the the t-statistic' is present. This repetition should be removed to maintain grammatical correctness.
- [**writing**] In the caption of Figure 9 (supplement.tex), the phrase 'the the t-statistic' appears. Please edit to remove the duplicate 'the' for clarity.
- [**writing**] In the caption of Figure 10 (supplement.tex), the phrase 'the the t-statistic' is repeated. Correct this typo to ensure the text reads smoothly.
- [**writing**] In the caption of Figure 11 (supplement.tex), the phrase 'the the t-statistic' is present. Please remove the extra 'the' to fix the grammatical error.
- [**writing**] In the caption of Figure 12 (supplement.tex), the phrase 'the the t-statistic' appears. This repetition should be corrected for better readability.
- [**writing**] In the caption of Figure 13 (supplement.tex), the phrase 'the the t-statistic' is repeated. Please edit to remove the duplicate 'the'.
- [**writing**] In the caption of Figure 14 (supplement.tex), the phrase 'the the t-statistic' is present. Correct this typo to ensure the text is grammatically sound.
- [**writing**] In the caption of Figure 15 (supplement.tex), the phrase 'the the t-statistic' appears. Please remove the extra 'the' to fix the grammatical error.
- [**writing**] In the caption of Figure 16 (supplement.tex), the phrase 'the the t-statistic' is repeated. This should be corrected for clarity and professionalism.
- [**writing**] In the caption of Figure 17 (supplement.tex), the phrase 'the the t-statistic' is present. Please edit to remove the duplicate 'the'.
- [**writing**] In the caption of Figure 18 (supplement.tex), the phrase 'the the t-statistic' appears. Correct this typo to ensure the text reads smoothly.
- [**writing**] In the caption of Figure 19 (supplement.tex), the phrase 'the the t-statistic' is repeated. Please remove the extra 'the' to fix the grammatical error.

- [**writing**] In the caption of Figure 20 (supplement.tex), the phrase 'the the t-statistic' is present. This repetition should be corrected for better readability.
- [**writing**] In the caption of Figure 21 (supplement.tex), the phrase 'the the t-statistic' appears. Please edit to remove the duplicate 'the'.
- [**writing**] In the caption of Figure 22 (supplement.tex), the phrase 'the the t-statistic' is repeated. Correct this typo to ensure the text is grammatically sound.
- [**writing**] In the caption of Figure 23 (supplement.tex), the phrase 'the the t-statistic' is present. Please remove the extra 'the' to fix the grammatical error.
- [**writing**] In the caption of Figure 24 (supplement.tex), the phrase 'the the t-statistic' appears. This repetition should be corrected for clarity and professionalism.
- [**writing**] In the caption of Figure 25 (supplement.tex), the phrase 'the the t-statistic' is repeated. Please edit to remove the duplicate 'the'.
- [**writing**] In the caption of Figure 26 (supplement.tex), the phrase 'the the t-statistic' is present. Correct this typo to ensure the text reads smoothly.
- [**writing**] In the caption of Figure 27 (supplement.tex), the phrase 'the the t-statistic' appears. Please remove the extra 'the' to fix the grammatical error.
- [**writing**] In the caption of Figure 28 (supplement.tex), the phrase 'the the t-statistic' is repeated. This should be corrected for better readability.
- [**writing**] In the caption of Figure 29 (supplement.tex), the phrase 'the the t-statistic' is present. Please edit to remove the duplicate 'the'.
- [**writing**] In the caption of Figure 30 (supplement.tex), the phrase 'the the t-statistic' appears. Correct this typo to ensure the text is grammatically sound.
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